

Topic 1 of Personnel Economics: The Principal-Agent Problem

Exercises

Exercise 1.1

Chapter 2 of Peter Kuhn (2018), *Personnel Economics*, discussion questions 2 and 3

Exercise 1.2

A company employs one worker. The worker has the following utility function:

$$U = w - \frac{1}{2}\theta e^2$$

where U is the utility of the worker, w is the worker's income, and e is the level of effort exerted by the worker. The parameter θ are both greater than zero. The worker aims to maximize utility. The worker's income w consists of the legal minimum wage w^* and a bonus per unit of output equal to b . The production function is $q=ke$. The income of the worker thus is $w=w^* + bke$. Throughout we assume that the worker is better off with this job than with any other job or without a job.

a) Derive the level of effort that the worker chooses.

Each unit of output yields revenues equal to p for the company. The company aims for maximum profits.

b) At what level of b does the firm maximize its profits?

c) Suppose the government decides to implement a tax on profits equal to $\tau\%$. The company continues and the worker remains at the company. The company aims for maximum after-tax profits. How does the profit-maximizing level of the bonus b change? Interpret your finding.

Exercise 1.3

Chapter 3 of Peter Kuhn (2018), *Personnel Economics*, discussion question 2.

Exercise 1.4 (copied (slightly adapted) from Peter Kuhn's teaching material)

Suppose the production function is $Q=E + k$, where $0 \leq k \leq 1$. The price of the product is equal to one. Note that k is an amount of production that is realized independent of the effort of the agent. The agent's utility function is $U = Y - E^2/2$ and the employment contract is $Y = bQ$. What is the profit-maximizing commission rate, b ? If $k = 0$, what is the profit-maximizing commission rate? What if $k=0.5$? $k=1$? Comment.

Exercise 1.5 (taken from the midterm of 2022)

A company employs one worker. The worker has the following utility function:

$$U = w + \gamma e - e^2$$

where U is the utility of the worker, w is the worker's income, e is the level of effort exerted by the worker, and γ is a positive parameter. The worker aims to maximize utility. The worker's income w consists of the legal minimum wage w^* and a bonus per unit of output equal to b . The production function is $q=e$. The income of the worker thus is $w=w^* + be$. Throughout we assume that the worker is better off with this job than with any other job or without a job.

a) Derive the effort level that maximizes the worker's utility.

b) Give one possible interpretation of the second term of the utility function (that is: γe). Write in plain words what this term may mean.

Profits of the company equal the production level q times the price of the product p minus the income of the worker minus energy costs:

$$\Pi = pq - w^* - bq - cq$$

where Π denotes profits and c denotes the energy cost per unit of output.

c) Derive the level of the bonus b that maximizes the company's profits. Explain in a clear sentence why the profit-maximizing bonus is lower when the energy cost c is higher.

Answers

1.1

Please find answers to Chapter 2, discussion questions 2 and 3, via this link:

<https://drive.google.com/open?id=1JKJfcX-rr0Oe2hENL9jpoZW8Z-KQdY8W>

1.2

a) Differentiating the worker's utility $w^* + b\kappa e - (1/2)\theta e^2$ to e and setting equal to zero gives the first-order condition for maximum utility: $b\kappa - \theta e = 0$. Solving for e gives $e = (b\kappa)/\theta$.

b) Profits equal $pq - w$. Substituting the production function and the definition of the wage gives: $p\kappa e - w^* - b\kappa e$. Substituting the effort level from the first part of this exercise $e = (b\kappa)/\theta$ gives $p\kappa(b\kappa/\theta) - w^* - b\kappa(b\kappa/\theta)$. Differentiating to b and setting equal to zero gives the first-order condition for maximum profits: $p\kappa(\kappa/\theta) - \kappa((b\kappa)/\theta) - b\kappa(\kappa/\theta) = 0$. Solving for b gives $b = ((p\kappa)/(2\kappa))$.

c) After-tax profits are $(1-\tau)[pq - w]$. Hence, maximizing after-tax profits amounts to maximizing before-tax profits. Hence, nothing changes to the profit-maximizing level of the bonus. This can also be verified by doing the analysis. When you take the derivative, the $(1-\tau)$ term drops out. The reason is that the tax applies equally to both the revenues and the costs. It reduces that marginal (after-tax) revenues. But also the marginal (after-tax) cost.

1.3

Please find the answer to Chapter 3, discussion question 2, via this link:

https://drive.google.com/open?id=1mB_8Z5mk2pE5F_xrD06l81CkWZtlqmcr

1.4 *(copied from Peter Kuhn's teaching material)*

To find the agent's optimal effort choice, substitute the contract and production functions into the agent's utility function to get $U = b(k+E) - E^2/2$. Maximizing this with respect E to for a given contract yields $E^* = b$, the same as in our baseline case. Profits can be written as $\Pi = (1-b)Q = (1-b)(E + k)$. Plugging E^* into this equation, we have $\Pi = (1-b)(b + k) = b - b^2 + k - bk$. Differentiating with respect to b gives the first-order condition $d\Pi/db = 1 - 2b - k = 0$, and re-arranging yields $b^* = (1-k)/2$.

Thus, when $k=0$ (the baseline case in the book), $b^*=0.5$. This is the case we worked out in Section 3.1. When $k = 0.5$, the profit-maximizing commission rate equals 0.25, and when $k=1$, the profit-maximizing commission rate is zero.

This example differs from the one in the book because, when $k > 0$, the agent produces some output ‘for free’, i.e. even when he exerts zero effort. One interpretation of this is intrinsic motivation: the agent likes doing the job so he’ll produce k units of output without incurring any effort costs. This changes the principal’s incentives because she now gets k units of output ‘for free’ without incentivizing the agent, i.e. even when $b = 0$. Raising b above zero lets the agent *keep* some of this free output, so the principal’s gains from raising b are smaller than in the base case when $k = 0$. This reduces the profit-maximizing value of b to below 0.5.

When $k = 1$ or higher, the agent produces so much ‘free’, unincentivized output that the principal’s best policy is not to incentivize the agent at all. An example might be a charitable organization with highly motivated volunteers, or an internship situation where the agent is so highly motivated by other factors (the ability to learn, make connections, and get a good recommendation) that zero cash incentives are optimal from the principal’s point of view.

Notice that this example contrasts with the “Laffer curve” reasoning underlying Result 3.1 in the text because we can now get positive output even with zero effort. Thus a 100% tax rate on the agent –i.e. $b = 0$, or zero financial rewards for working—*can* be optimal now.

1.5

a) The worker’s utility can be written as $U = w^* + be + \gamma e - e^2$. Taking the derivative to e and setting equal to zero gives:

$$b + \gamma - 2e = 0$$

Solving yields $e = (b + \gamma)/2$.

b) The term may represent the worker’s intrinsic motivation, the worker’s joy from exerting effort, or the worker’s care about the output of the company. (One of these answers (or another one that makes sense) suffices.)

c) Using the production function $q = e$ and the optimum effort derived under a), profits can be written as: $\Pi = (p - b - c)(b + \gamma)/2 - w^*$. Taking the derivative to b , setting equal to zero, and solving gives: $b = (p - \gamma - c)/2$. Clearly, when the energy cost c is higher, the company benefits less from increasing output, as it needs to pay a higher energy cost c for each unit of output. Hence, it sets a lower bonus.